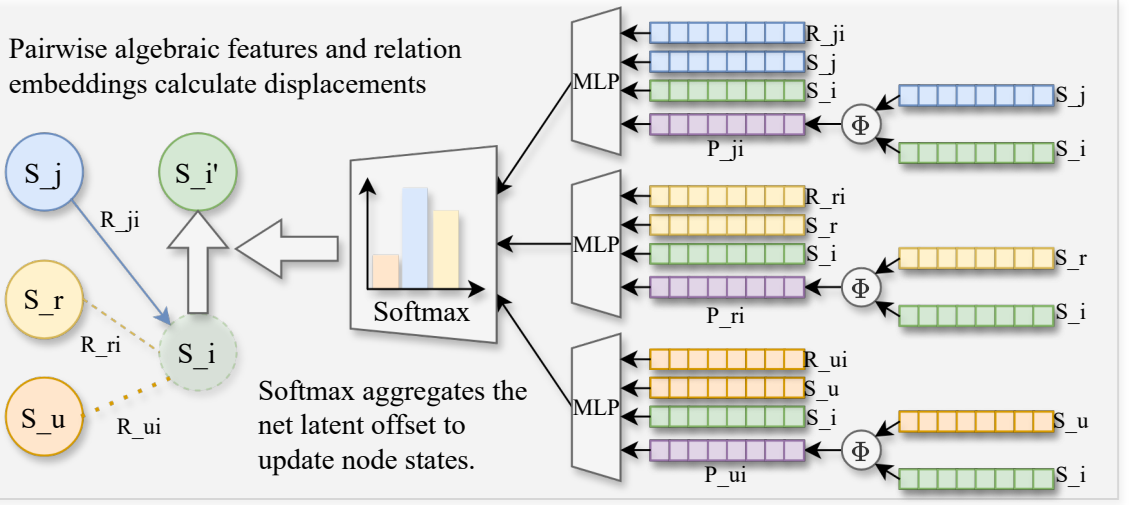

Pairwise algebraic features and relation embeddings calculate displacements
S_j
S_i'
R_ji
S_r
R_ri
S_i
S_u
R_ui
Softmax
Softmax aggregates the net latent offset to update node states.
MLP
R_ji
S_j
S_i
P_ji
Φ
S_j
S_i
MLP
R_ri
S_r
S_i
P_ri
Φ
S_r
S_i
MLP
R_ui
S_u
S_i
P_ui
Φ
S_u
S_i